Manuscript Correction Summary

Project 1 - Spatial CRT

Andrew Walther

2026-11-04

Summary

While reviewing the paper and corresponding code for thesis integration, we found two coding errors that affect only the 3×3 and 3×4 grids. The summary script used the 2×4 BSS allocations for every grid, such that the BSS columns for 3×3 and 3×4 summarized two allocations that are not the block stratified design on those grids. The control-only spillover loop in the 3×3 and 3×4 simulation scripts also stopped at district 8 (for the 2x4 grid) instead of indexing to 9 and 12. Therefore, the higher-numbered (9-12) treated districts never passed spillover to their control neighbors. That second error changed the SRS control-only results only.

Fixing both issues changes two figures (Figures 6 and 7), two tables (Tables 2 and 3), the results paragraphs for the 3×3 and 3×4 grids, and one paragraph of the discussion section (the "checkerboard paradox", which is removed). A handful of summary sentences elsewhere are altered to match the updated results. The methods, the 2×4 results (Table 1, Figure 5) and the North Carolina recommendation (Application) do not change.

The paper's findings and its recommendation are unchanged, and the corrected results support them more consistently across grids than the published ones did. After the fix, the 3×4 grid follows the same pattern as the 2×4 grid in all 16 scenarios. On every grid SRS has the lower average MSE when spillover reaches both arms, BSS has the lower MSE when spillover goes into control clusters and spatial dependence is present, and BSS is usually the less variable design. Some individual 3×3 and 3×4 values do move a lot, and in two settings the better design flips. The tables and figures below show each of those changes.

Findings and recommendation

Design with the lower mean MSE across the four ψ values in each setting. Bold marks a winner that differs from the 2×4 grid. The last column counts scenarios where the SD of MSE across allocations is smaller for BSS.

	Control Only		Intervention & Control		
Grid	$\rho = 0.01$	$\rho = 0$	$\rho = 0.01$	$\rho = 0$	BSS less variable
2×4 (unchanged)	BSS 4/4	SRS 4/4	SRS 4/4	SRS 4/4	16/16
3×3 before	BSS 4/4	**BSS 4/4**	**BSS 4/4**	**BSS 4/4**	16/16
3×3 after	BSS 4/4	**BSS 4/4**	SRS 4/4	SRS 4/4	8/16
3×4 before	BSS 4/4	**BSS 4/4**	SRS 4/4	**BSS 4/4**	12/16
3×4 after	BSS 4/4	SRS 4/4	SRS 4/4	SRS 4/4	14/16

Prior to the correction, the 3×3 grid showed BSS leading SRS everywhere, and the 3×4 grid showed BSS leading in every setting except intervention-and-control spillover with $\rho = 0.01$. Those results were derived from incorrect allocations. After the fix, the only disagreement between grids is control-only spillover with $\rho = 0$, where BSS still outperforms SRS on the 3×3 grid.

The paper's claims, checked against the corrected results.

Claim in the accepted paper	After the correction
SRS vs. BSS is a risk-management trade-off (BSS stable, SRS better on average but variable)	Holds. Exception is 3×3 intervention-and-control spillover, where BSS is neither lower nor more stable
BSS preferred when spillover flows into controls, especially with spatial dependence	Holds, and now consistent across all three grids
SRS often has lower MSE when spillover reaches both arms	Holds, and stronger. SRS wins every such scenario on every grid (published 3×3 said the opposite)
MSE rises with ψ	Holds
Spatial dependence raises MSE for both designs	Holds for SRS on every grid. For BSS only on 3×3 (it was already false for 2×4 BSS as published)

The paper's claims, checked against the corrected results. *(continued)*

Claim in the accepted paper	After the correction
SRS has lower bias, BSS lower variance	Holds for 2×4 and 3×4, not for 3×3
BSS "consistently exhibits minimal MSE"	Dropped. Not true on any grid, including the unchanged 2×4
7 allocations are each optimal somewhere, BSS never the single best	BSS never best still holds. Counts are now 7, 4 and 6 (was 7, 7, 7)
"Checkerboard paradox" (BSS backfires under bidirectional spillover with $\rho > 0$)	Removed. The real mechanism is treatment-spillover collinearity in checkerboards (bias near $-\psi$)
NC probation recommendation (BSS when spillover flows mainly to controls)	Holds

Figures 6 and 7

In each pair the published figure is on the left and the corrected one on the right.

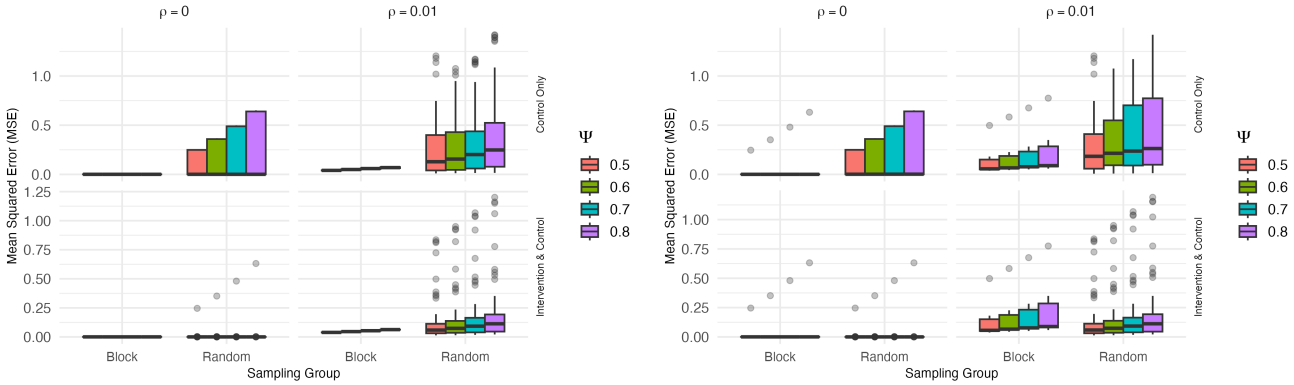


Figure 6, 3×3 grid. The BSS boxes are now built from the six valid 3×3 allocations instead of the BSS allocations from the 2x4 grid. Under $\rho = 0.01$ they rise from near zero to a level below SRS for control-only spillover but above SRS for intervention-and-control spillover. Under $\rho = 0$ most BSS allocations still sit near zero, with one outlier (combo 77, a checkerboard-like allocation where $\widehat{\beta}$ absorbs the spillover). The SRS control-only boxes rise slightly from the loop fix.

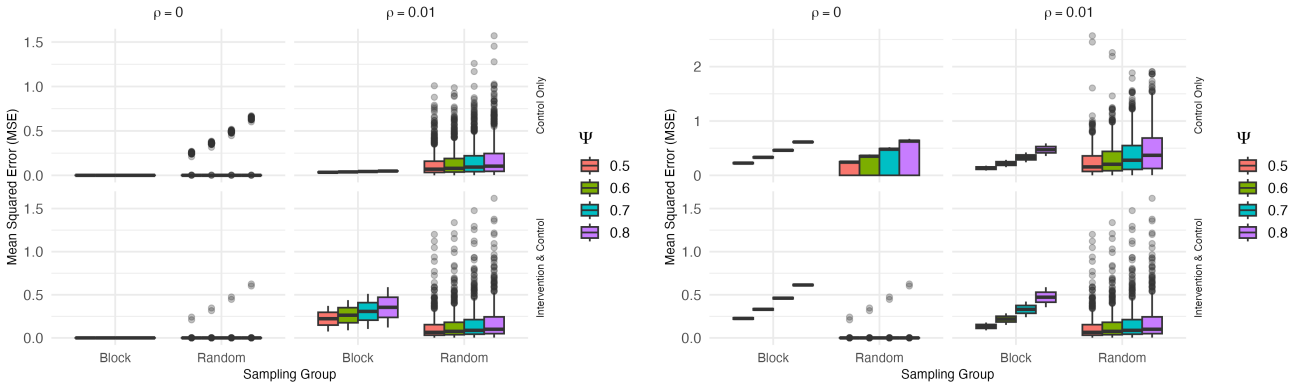


Figure 7, 3×4 grid. Now BSS is the two 3×4 checkerboards (combos 313 and 612), which give identical results under both spillover types. Their MSE rises with ψ and sits well above zero in every panel. The SRS control-only boxes roughly double after the loop fix (y-axis now runs past MSE=2). The intervention-and-control SRS panels are unchanged.

Tables 2 and 3

Each cell shows the corrected value. When a value changed, the published value comes first and the cell is shaded. Units follow the mechanics of the original paper.

Manuscript Table 2 (3×3 grid), published → corrected. 75 of 96 cells changed.

	Simple Random Sampling			Block Stratified Sampling		
ψ	MSE	SD (×10)	Bias	MSE	SD (×10)	Bias
Control Only, $\rho = 0.01$						
0.5	0.234 → 0.274	2.593 → 2.631	-0.054 → -0.089	0.041 → 0.147	0.044 → 1.799	-0.012 → -0.054
0.6	0.276 → 0.330	2.860 → 2.964	-0.092 → -0.135	0.050 → 0.174	0.080 → 2.115	-0.018 → -0.064
0.7	0.322 → 0.390	3.284 → 3.472	-0.131 → -0.181	0.059 → 0.205	0.114 → 2.460	-0.022 → -0.073
0.8	0.372 → 0.455	3.857 → 4.140	-0.170 → -0.227	0.069 → 0.240	0.145 → 2.839	-0.025 → -0.083
Control Only, $\rho = 0$						
0.5	0.092 → 0.114	1.207 → 1.247	-0.182 → -0.226	0.001 → 0.042	0.002 → 1.002	0.003 → -0.078
0.6	0.132 → 0.164	1.740 → 1.798	-0.219 → -0.271	0.001 → 0.059	0.002 → 1.434	0.003 → -0.096
0.7	0.180 → 0.223	2.370 → 2.449	-0.255 → -0.316	0.001 → 0.081	0.002 → 1.959	0.003 → -0.113
0.8	0.235 → 0.291	3.099 → 3.203	-0.292 → -0.362	0.001 → 0.106	0.002 → 2.577	0.003 → -0.130
Intervention & Control, $\rho = 0.01$						
0.5	0.103	1.480	0.071	0.039 → 0.147	0.022 → 1.799	-0.019 → -0.054
0.6	0.122	1.688	0.082	0.046 → 0.174	0.027 → 2.115	-0.020 → -0.064
0.7	0.143	1.905	0.092	0.053 → 0.205	0.032 → 2.460	-0.021 → -0.073
0.8	0.168 → 0.167	2.149 → 2.133	0.101 → 0.102	0.063 → 0.240	0.010 → 2.839	-0.019 → -0.083
Intervention & Control, $\rho = 0$						
0.5	0.002	0.219	-0.004	0.001 → 0.042	0.004 → 1.002	0.004 → -0.078
0.6	0.003	0.313	-0.004	0.001 → 0.059	0.004 → 1.434	0.004 → -0.096
0.7	0.004	0.428	-0.005	0.001 → 0.081	0.004 → 1.959	0.004 → -0.113
0.8	0.005	0.562	-0.006	0.001 → 0.106	0.004 → 2.577	0.004 → -0.130

In Table 2, the published BSS values of about 0.001 under $\rho = 0$ come from two allocations that are not block stratified on 3×3. For the six valid allocations, BSS MSE is 0.042 to 0.106 and still well below SRS for control-only spillover, but above SRS for intervention-and-control spillover. The one SRS change under intervention-and-control spillover ($\psi = 0.8$, $\rho = 0.01$) comes from regenerating a stored result file and only the third decimal place is altered.

Manuscript Table 3 (3×4 grid), published → corrected. 72 of 96 cells changed.

	Simple Random Sampling			Block Stratified Sampling		
ψ	MSE	SD (×100)	Bias	MSE (×10)	SD (×100)	Bias
Control Only, $\rho = 0.01$						
0.5	0.120 → 0.244	13.244 → 24.422	0.033 → -0.190	0.346 → 1.332	0.963 → 6.416	-0.152 → -0.273
0.6	0.138 → 0.311	14.807 → 28.926	0.021 → -0.242	0.388 → 2.180	0.505 → 9.471	-0.167 → -0.391
0.7	0.159 → 0.390	17.166 → 34.937	0.009 → -0.292	0.434 → 3.309	0.011 → 12.806	-0.180 → -0.509
0.8	0.183 → 0.480	20.387 → 42.507	-0.001 → -0.341	0.484 → 4.721	0.561 → 16.429	-0.193 → -0.627
Control Only, $\rho = 0$						
0.5	0.034 → 0.133	8.554 → 12.497	-0.068 → -0.264	0.004 → 2.255	0.022 → 2.274	-0.006 → -0.463
0.6	0.049 → 0.191	12.318 → 18.005	-0.081 → -0.317	0.004 → 3.315	0.022 → 2.196	-0.006 → -0.570
0.7	0.067 → 0.260	16.769 → 24.518	-0.094 → -0.370	0.004 → 4.609	0.020 → 1.991	-0.006 → -0.676
0.8	0.087 → 0.339	21.912 → 32.042	-0.108 → -0.423	0.003 → 6.137	0.016 → 1.659	-0.006 → -0.782
Intervention & Control, $\rho = 0.01$						
0.5	0.112	12.873	0.022	2.229 → 1.332	20.995 → 6.416	-0.415 → -0.273
0.6	0.131	14.762	0.021	2.634 → 2.180	24.750 → 9.471	-0.452 → -0.391
0.7	0.151	16.817	0.020	3.074 → 3.309	28.774 → 12.806	-0.488 → -0.509
0.8	0.173	19.046	0.018	3.548 → 4.721	33.060 → 16.429	-0.525 → -0.627
Intervention & Control, $\rho = 0$						
0.5	0.001	1.051	-0.001	0.003 → 2.255	0.004 → 2.274	-0.003 → -0.463
0.6	0.001	1.542	-0.001	0.003 → 3.315	0.004 → 2.196	-0.002 → -0.570
0.7	0.001	2.143	-0.002	0.003 → 4.609	0.005 → 1.991	-0.002 → -0.676
0.8	0.002	2.853	-0.002	0.003 → 6.137	0.005 → 1.659	-0.002 → -0.782

In Table 3, the BSS MSE (×10 in the table) increases from near zero to 0.226 to 0.614 on the original scale under $\rho = 0$. This is due to the fact that in a BSS allocation the spillover indicator equals 1 − x and $\hat{\beta}$ absorbs the spillover. The BSS bias is close to $-\psi$ there (-0.463 at $\psi = 0.5$). The SRS control-only MSE approximately doubles, and its mean bias goes from slightly positive to clearly negative (0.033 to -0.190 at $\psi = 0.5$, $\rho = 0.01$). SRS under intervention-and-control spillover is unchanged.

Text changes

The text edits are limited to statements that depended on the 3×3 and 3×4 numbers. Section names follow the manuscript.

Text changes, by manuscript section. CO = control-only spillover, IC = intervention-and-control spillover.

Section	Before	After
Abstract	BSS gives "minimal" or "near-zero" variation in MSE and "minimizes" variance.	Wording softened to "generally less" and "generally reduces". No numbers in the abstract change.
Results, 2×4 grid	Spatial dependence raises MSE for both designs on every grid.	True for SRS everywhere. For BSS on 2×4, $\rho = 0.01$ lowers MSE (0.165 vs. 0.262 at $\psi = 0.5$). Results unchanged, sentence corrected.
Results, 3×3 grid	BSS beats SRS on MSE, SD and bias under both spillover types. BSS MSE about 0.001 (CO, $\rho = 0$) and 0.039 (IC, $\rho = 0.01$).	BSS still wins every control-only scenario (MSE 0.042 to 0.106 vs. SRS 0.114 to 0.291 at $\rho = 0$). Under both-arm spillover SRS wins every scenario (BSS 0.147 to 0.240, SRS 0.103 to 0.167 at $\rho = 0.01$).
Results, 3×4 grid	BSS "dramatically outperforms" SRS under control-only spillover (0.0004 vs. 0.034 at $\rho = 0$). Under both-arm spillover with $\rho = 0.01$, BSS loses through a "checkerboard paradox".	Same pattern as 2×4 in every scenario. Control-only, BSS wins at $\rho = 0.01$ (0.133 vs. 0.244) and SRS at $\rho = 0$ (0.133 vs. 0.226). Both-arm spillover, SRS wins everywhere. Paradox replaced by the collinearity explanation.
Recommendations	If bidirectional spillover and spatial dependence are expected, BSS can amplify bias (checkerboard paradox).	If spillover among treated units is expected, SRS had lower average MSE in every such scenario. The recommendation itself is unchanged.
Discussion, bias and variance	SRS gives lower bias, BSS "consistently" lower variance.	Holds on 2×4 and 3×4. On 3×3 BSS has the smaller bias in most scenarios and varies more under both-arm spillover.
Discussion, effect of ρ	Even small ρ generally raises MSE for both designs.	Always raises MSE for SRS. For BSS it rises on 3×3 but falls for the 2×4 and 3×4 checkerboards.
Discussion, spillover types	Control-only spillover favors BSS. Both-arm spillover often favors SRS.	Same message, made exact. Control-only with $\rho > 0$ favors BSS on every grid. Both-arm spillover favors SRS in every scenario.
Discussion, optimal allocations	7 allocations are each optimal on every grid. BSS "consistently exhibits minimal MSE", so it is prudent to recommend.	Counts are 7, 4 and 6. BSS is never the single best and does not give the lowest average MSE. Its value is limiting the worst case.
Discussion, checkerboard paragraph	Names the "checkerboard paradox": complete separation of arms amplifies spillover bias under bidirectional spillover.	Replaced by a short note that in a checkerboard spillover equals 1 − x under both spillover types, so β and ψ are not separately identifiable and $\hat{\beta}$ has bias near $-\psi$.
Limitations, Future Considerations	Collinear BSS error is bias because the BSS SD is near zero. Future work targets the "amplified bias" scenarios.	Reason restated as variance small relative to squared bias. Future work targets allocations where treatment and spillover are collinear.
Declarations	"National Institutes of Mental Health".	"National Institute of Mental Health" (typo fix, unrelated to the correction).